

Functional Requirement Document

Here is the detailed Functional Requirement Document (FRD) based on the provided Business Requirement Document (BRD) and adhering to the "Coding Requirements" section.

FR-DLT-001

- Title: Load Customer and Order Data into Delta Tables
- Description: The system must load customer and order data from CSV files into delta tables, perform data cleansing, and create aggregated tables.
- Preconditions:
 - CSV files are available at the specified locations:
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` and
`/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata`
 - The delta tables do not exist or are empty
- Main Flow / Functional Steps:
 - Read customer data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/customerdata` into a delta table named "customer" with schema: CustId, Name, EmailId, Region
 - Read order data from CSV file `/Volumes/gen\_ai\_poc\_databrickscoe/sdlc\_wizard/orderdata` into a delta table named "order" with schema: OrderId, ItemName, PricePerUnit, Qty, Date, CustId
 - Add a new column "TotalAmount" to the "order" delta table, calculated as the product of "PricePerUnit" and "Qty"
 - Remove null and duplicate records from both "customer" and "order" delta tables
 - Create a new delta table "ordersummary" in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist, with schema: CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date
 - Join "customer" and "order" delta tables on "CustId" and load the joined data into the "ordersummary" delta table as an SCD type 2 table
 - Create a new delta table "customeraggregatespend" in catalog "gen_ai_poc_databrickscoe" and schema "sdlc_wizard" if it does not exist, with schema: Name, TotalAmount, Date
 - Aggregate the "TotalAmount" column from the "ordersummary" delta table, grouped by "Name" and "Date", and load the aggregated data into the "customeraggregatespend" delta table
 - Implement SCD type 2 logic to update the "ordersummary" delta table when there are changes in the "customer" delta table, including:
 - Marking old records as inactive
 - Creating new records as active
 - Updating StartDate and EndDate accordingly

FR-DLT-002

- Title: Implement Delta Live Tables (DLT) for Data Processing
- Description: The system must implement Delta Live Tables (DLT) to automate the data processing pipeline.
- Preconditions:
 - Delta tables are created and populated as per FR-DLT-001
- Main Flow / Functional Steps:
 - Create a DLT pipeline that automates the data processing steps outlined in FR-DLT-001
 - Use DLT to read data from CSV files, perform data cleansing, and create aggregated tables
 - Use DLT to implement SCD type 2 logic for the "ordersummary" delta table

Let me know if this meets the requirements or if you need further modifications!